

WORKSHEET: CHEMICAL FORMULAE

1. **Without looking at your notes**, write the symbol and valency for the following ions. The more you can commit to memory, the easier it will be to write formulae correctly.

potassium ion K^{1+}

nitrate ion $(NO_3)^{1-}$

magnesium ion Mg^{2+}

calcium ion Ca^{2+}

hydroxide ion $(OH)^{1-}$

sulphate ion $(SO_4)^{2-}$

aluminium ion Al^{3+}

oxide ion O^{2-}

chloride ion Cl^{1-}

sodium ion Na^{1+}

silver ion Ag^{1+}

carbonate ion $(CO_3)^{2-}$

phosphate ion $(PO_4)^{3-}$

sulphide ion S^{2-}

zinc ion Zn^{2+}

iron (III) ion Fe^{3+}

bromide ion Br^{1-}

copper (II) ion Cu^{2+}

2. Write the names for the following compounds. *Note - some of them are not ionic!*



magnesium chloride



potassium nitrate



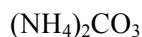
barium sulphate



nitrogen dioxide



zinc nitride



ammonium carbonate



potassium phosphate



dinitrogen tetroxide



diphosphorus pentoxide



carbon monoxide



iron (III) hydroxide



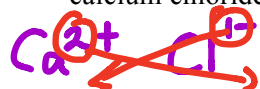
sodium sulphide

3. Write the formulae for the following compounds. Show the ions you are using - it helps!

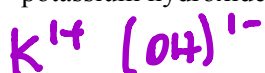
sodium chloride



calcium chloride



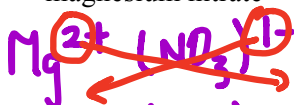
potassium hydroxide



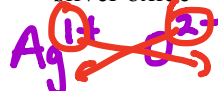
potassium sulphide



magnesium nitrate



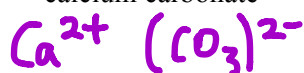
silver oxide



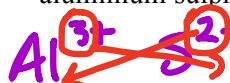
aluminium chloride



calcium carbonate



aluminium sulphide



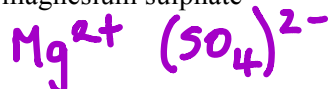
sodium carbonate



aluminium oxide



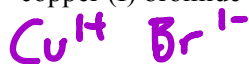
magnesium sulphate



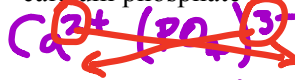
aluminium sulphate



copper (I) bromide



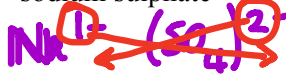
calcium phosphate



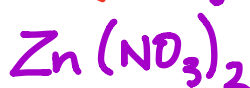
iron (III) sulphide



sodium sulphate



zinc nitrate



zinc bromide



magnesium phosphate

